

Experimental Validation of 3D Visible Light Positioning Using a Single Beam-Steered LED

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Abstract—This paper presents, to the best of our knowledge, the first experimental validation of three-dimensional visible-light positioning based on a single beam-steered LED (SBL) and a single photodiode receiver. The method builds on a previously developed ratio-based formulation, in which inter-orientation RSS ratios cancel the receiver-orientation term, and evaluates two closed-form estimators, namely generalized least squares (GLS) and weighted least squares (WLS), under a $K = 9$ steering configuration. The experimental campaign was conducted on a 3D testbed comprising a commercial IR LED mounted on a two-axis motorized gimbal and a photodiode. The results show that the SBL principle is experimentally feasible in 3D with a minimal receiver architecture, and that GLS consistently outperforms WLS. In particular, GLS achieves a 3D positioning RMSE of 12.52 cm, compared with 15.79 cm for WLS. In terms of direction finding, GLS yields a pointing-error RMSE of 4.87° , confirming its superior robustness, especially for large off-axis inclinations. Finally, an integration-time analysis shows that the localization accuracy remains essentially stable over a wide range of averaging windows, while the estimate precision improves rapidly with modest temporal averaging. Overall, the results validate the practical viability of single-LED 3D SBL positioning and highlight its relevance for beam-aware indoor optical wireless systems.

Index Terms—Optical wireless positioning, 3D indoor localization, beam steering.

I. INTRODUCTION

Indoor positioning is a key enabler for emerging applications such as robotics, asset tracking, navigation, and context-aware wireless services. In this context, visible light positioning (VLP), and more broadly optical wireless positioning (OWP), have attracted significant interest because they can exploit LED-based infrastructure while offering high spatial reuse, low electromagnetic interference, and direct integration with optical wireless communication (OWC) systems [1], [2]. In particular, photodiode (PD)-based VLP remains attractive for practical deployment due to its low hardware complexity and high sampling capability. However, most PD-based VLP systems still follow a multi-anchor paradigm, in which several ceiling luminaires must be simultaneously visible to the user in order to enable robust positioning [3]. This requirement increases installation and calibration complexity and may also reduce effective coverage in sparsely lit areas or under field-of-view and blockage constraints, thereby motivating architectures that can reduce the dependence on dense transmitter deployments.

To mitigate infrastructure density, recent research has increasingly investigated single-LED positioning. Existing single-LED demonstrations typically obtain the missing geometric degrees of freedom by increasing receiver-side capability, e.g., using image sensors and projective geometry [4]–[6], employing multi-photodiode (multi-PD) receivers with tilted elements [7], or introducing receiver rotation, tracking, and/or learning-based inference [8], [9]. While these approaches validate the feasibility of single-LED localization, they often shift complexity to the user equipment (UE), through camera sensing, inertial aiding, mechanical actuation, or data-driven training pipelines, which may reduce interpretability and complicate generalization across devices and environments [9].

In parallel, agile optical beam steering is becoming a core enabler in indoor optical wireless communication (OWC), supporting high-capacity links, dynamic beam management, and localization functionalities based on controlled beam scanning [10]. This naturally suggests a different architectural path for single-LED positioning: instead of increasing UE complexity, diversity can be generated at the infrastructure side using a single beam-steered LED (SBL) that probes the receiver from multiple controlled orientations.

Motivated by this, our recent theoretical work in [11] introduced a ratio-based SBL framework that enables 3D localization with a single PD by steering the transmitter over a set of orientations and exploiting inter-orientation RSS ratios to eliminate the receiver-orientation term under the ideal model. The resulting direction-finding problem admits statistically grounded closed-form solutions, including generalized least squares (GLS) and a lower-complexity weighted least squares (WLS) approximation. A related 2D beam-steering formulation with experimental tilt-robustness validation was reported in [12].

This model-based viewpoint is also important from a methodological perspective. While machine-learning-based VLP has received growing attention and can improve performance in complex environments, it typically relies on training data and implicit nonlinear mappings whose behavior may be difficult to interpret physically and to generalize across devices or deployment conditions [9]. By contrast, the SBL formulation considered here is not a black box: the estimators are derived directly from radiometric geometry, preserve a closed-form structure, and allow the impact of steering, noise, and ratio uncertainty to be explicitly analyzed. However,

despite these theoretical advances, the experimental validation of a 3D single beam-steered LED / single-PD positioning architecture has remained open.

This paper presents, to the best of our knowledge, the first experimental validation of the above 3D SBL principle using a single beam-steered LED and a single PD receiver on a 3D testbed. Specifically, we (i) validate the feasibility of 3D positioning with a single steerable LED in real hardware, (ii) experimentally compare GLS and WLS for both 3D positioning and direction finding, and (iii) study positioning stability as a function of RSS integration time, highlighting the accuracy–latency trade-off relevant to practical SBL operation.

The remainder of the paper is organized as follows. Section II summarizes the SBL system model and the GLS/WLS estimators. Section III describes the experimental testbed and measurement protocol. Section IV reports the experimental results in terms of 3D positioning accuracy, direction finding performance, and estimation stability. Section V concludes the paper.

II. SYSTEM MODEL AND ESTIMATORS

This section summarizes the ideal single beam-steered LED (SBL) model adopted in this work and the two closed-form direction estimators considered in the experimental evaluation, namely generalized least squares (GLS) and weighted least squares (WLS). Full derivations are given in [11]; here, we retain only the key equations required to support the remainder of the paper.

A. Theoretical Ratio-Based SBL Positioning

Consider a single LED transmitter located at the known ceiling position $\mathbf{p}_t = [0, 0, H]^\top$ and a photodiode (PD) receiver located at the unknown position $\mathbf{p}_r = [x_r, y_r, z_r]^\top$. The LED sequentially steers its boresight over a set of K predefined orientations, indexed by $k \in \{1, \dots, K\}$, with corresponding unit vectors $\mathbf{n}_{t,k}$. The received signal strength (RSS) for the single-LED case follows the general LOS optical model [13]:

$$S_k = R_{\text{pd}} P_t R(\phi_k) A_{\text{eff}}(\psi_k) / d^2, \quad (1)$$

where R_{pd} is the PD responsivity, P_t is the emitted optical power, $R(\phi_k)$ is the transmitter radiation pattern, $A_{\text{eff}}(\psi_k)$ is the effective PD area at incidence angle ψ_k , and d is the LOS distance. For the SBL configuration considered here, k indexes the steering orientations.

Under the ideal Lambertian assumption adopted in [11], (1) becomes:

$$S_k = C \cos^m(\phi_k) \cos(\psi_k) / d^2, \quad (2)$$

with:

$$C = R_{\text{pd}} P_t \frac{(m+1) A_{\text{det}}}{2\pi}, \quad (3)$$

where m denotes the Lambertian order. Defining the displacement vector $\mathbf{d} = \mathbf{p}_r - \mathbf{p}_t$ and the corresponding unit direction vector $\mathbf{n}_d = \mathbf{d} / \|\mathbf{d}\|$, the irradiance and incidence cosines are:

$$\cos(\phi_k) = \mathbf{n}_{t,k} \cdot \mathbf{n}_d, \quad \cos(\psi) = \mathbf{n}_r \cdot (-\mathbf{n}_d). \quad (4)$$

The key property of the SBL geometry is that, for a fixed receiver position, the factor $\cos(\psi)$ is common to all steering orientations. Therefore, inter-orientation RSS ratios cancel the receiver-orientation term exactly under the ideal model. Using orientation $k = 1$ as reference, the theoretical ratio is defined as:

$$\beta_k = \left(\frac{S_k}{S_1} \right)^{\frac{1}{m}} = \frac{\cos(\phi_k)}{\cos(\phi_1)} = \frac{\mathbf{n}_{t,k} \cdot \mathbf{n}_d}{\mathbf{n}_{t,1} \cdot \mathbf{n}_d}, \quad k = 2, \dots, K, \quad (5)$$

Hence, each ratio induces the homogeneous constraint:

$$\mathbf{a}_k^\top \mathbf{n}_d = 0, \quad \mathbf{a}_k = \mathbf{n}_{t,k} - \beta_k \mathbf{n}_{t,1}, \quad (6)$$

showing that the unknown direction vector lies at the intersection of the ratio-defined hyperplanes.

In practice, the direction-finding stage is first solved from the K ratio measurements; a subsequent range-recovery step can then reconstruct the 3D receiver position as detailed in [11]. In this paper, the same optimized SBL configuration with $K = 9$ steering orientations is adopted throughout the experimental study.

B. GLS and WLS

At each steering orientation, the receiver acquires N RSS samples. Their sample mean is $\bar{S}_k = \frac{1}{N} \sum_{n=1}^N S_k[n]$, for $k = 1, \dots, K$. Replacing the noiseless RSS values in (5) by the sample means yields the empirical ratios:

$$\hat{\beta}_k = \left(\frac{\bar{S}_k}{\bar{S}_1} \right)^{1/m}, \quad k = 2, \dots, K, \quad (7)$$

and the associated empirical constraint vectors:

$$\hat{\mathbf{a}}_k = \mathbf{n}_{t,k} - \hat{\beta}_k \mathbf{n}_{t,1}. \quad (8)$$

Stacking them columnwise gives:

$$\hat{\mathbf{A}} = [\hat{\mathbf{a}}_2, \dots, \hat{\mathbf{a}}_K] \in \mathbb{R}^{3 \times (K-1)}. \quad (9)$$

Because all empirical ratios share the same reference measurement $\bar{S}_1$, the induced ratio errors are heteroscedastic and mutually correlated. Let Σ_β denote the closed-form covariance matrix of these ratio errors, derived in [11]. The GLS estimator accounts for this full covariance and computes the direction as:

$$\hat{\mathbf{n}}_{d,\text{GLS}} = \arg \min_{\|\mathbf{u}\|=1} \mathbf{u}^\top \left(\hat{\mathbf{A}} \Sigma_\beta^{-1} \hat{\mathbf{A}}^\top \right) \mathbf{u}. \quad (10)$$

Since (10) is a constrained Rayleigh quotient, its closed-form solution is the unit eigenvector associated with the smallest eigenvalue of:

$$\mathbf{M}_{\text{GLS}} = \hat{\mathbf{A}} \Sigma_\beta^{-1} \hat{\mathbf{A}}^\top, \quad (11)$$

namely:

$$\hat{\mathbf{n}}_{d,\text{GLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\text{GLS}})}{\|\mathbf{v}_{\min}(\mathbf{M}_{\text{GLS}})\|}, \quad \hat{\mathbf{n}}_{d,\text{GLS}} \cdot \mathbf{n}_{t,1} > 0, \quad (12)$$

where the sign condition enforces a physically consistent direction from transmitter to receiver.

A lower-complexity alternative is obtained by neglecting the off-diagonal terms of Σ_β and retaining only the ratio variances. Defining:

$$\tilde{\Sigma}_\beta = \text{diag}(\Sigma_\beta), \quad (13)$$

the WLS estimator is:

$$\hat{\mathbf{n}}_{d,\text{WLS}} = \arg \min_{\|\mathbf{u}\|=1} \mathbf{u}^\top \left(\hat{\mathbf{A}} \tilde{\Sigma}_\beta^{-1} \hat{\mathbf{A}}^\top \right) \mathbf{u}, \quad (14)$$

whose closed-form solution is:

$$\hat{\mathbf{n}}_{d,\text{WLS}} = \frac{\mathbf{v}_{\min}(\mathbf{M}_{\text{WLS}})}{\|\mathbf{v}_{\min}(\mathbf{M}_{\text{WLS}})\|}, \quad \mathbf{M}_{\text{WLS}} = \hat{\mathbf{A}} \tilde{\Sigma}_\beta^{-1} \hat{\mathbf{A}}^\top. \quad (15)$$

Therefore, both estimators preserve the same model-based closed-form structure, while differing in how they weight the ratio-induced constraints: GLS uses the full covariance model and is statistically more accurate, whereas WLS provides a simpler approximation by ignoring cross-correlation terms. Their experimental comparison is presented in Section IV.

III. EXPERIMENTAL TESTBED

A. 3D Testbed Configuration

The experimental validation of the proposed 3D SBL-VLP framework was conducted on a testbed covering a volume of $2.8 \text{ m} \times 2.8 \text{ m} \times 2.0 \text{ m}$. The transmitter was installed at the ceiling center, located at $\mathbf{p}_t = [0, 0, 2]^\top \text{ m}$, whereas the receiver workspace was defined within a region of interest (RoI) spanning heights from 0.4 m to 1.2 m. A regular 3D grid with a spacing of 0.2 m yields a total of 1125 candidate sampling locations, from which 300 representative 3D positions were selected for the present study.

The SBL transmitter consists of a commercial IR LED (LZ1-00R702) driven by a PT4115 current driver supplying a controlled current of 0.77 A, which corresponds to an optical power of approximately $P_t = 1 \text{ W}$. The LED was mounted on a motorized two-axis gimbal that emulates beam steering. To approximate pure rotational steering with negligible translational displacement of the source, the LED was mechanically aligned with the intersection of the two gimbal rotation axes. The gimbal was implemented using two stacked Thorlabs PRM1Z8 motorized rotary stages, each with an angular resolution of 0.1° , mounted on an aluminum-profile structure.

At the receiver side, a silicon PIN photodiode (BPX61) was mounted on a custom PCB comprising a transimpedance amplifier followed by a voltage amplification stage. Both stages were implemented using OP27 operational amplifiers and powered by a dual $\pm 15 \text{ V}$ supply. The receiver module was attached to the end-effector of a 6-DOF Universal Robots UR5 collaborative arm, which provided accurate and repeatable positioning over the 3D RoI. The analog output of the receiver front-end was acquired by a NI USB-6421 DAQ device operating at a sampling rate of 1 kHz. A photograph of the complete experimental setup is shown in Fig. 1.

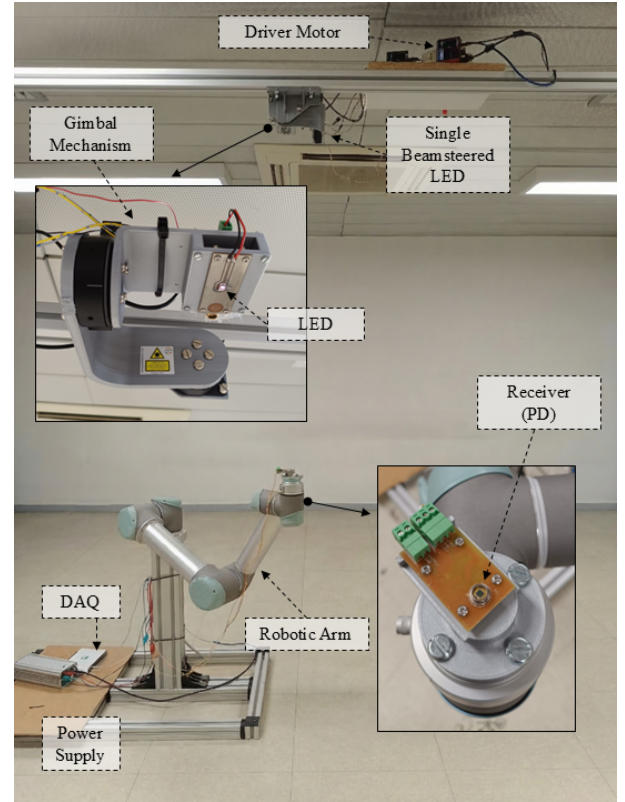


Fig. 1. Photograph of the 3D robotic SBL-VLP testbed. The transmitter consists of a commercial IR LED mounted at the center of a two-axis motorized gimbal, while the receiver is a BPX61 photodiode front-end attached to the end-effector of a UR5 robotic arm for precise 3D positioning.

B. Experimental Protocol for the SBL Method Evaluation

For each receiver location, the measurement procedure was carried out as follows. First, the robotic arm placed the photodiode at the desired ground-truth 3D coordinate. Then, with the LED continuously turned on, the gimbal sequentially steered the transmitter toward the $K = 9$ predefined orientations of the SBL configuration under test. For each orientation, the DAQ recorded 10^3 voltage samples, corresponding to 1 s of observation time. The recorded voltage samples were stored as the raw RSS observations associated with the corresponding steering direction and 3D receiver position. This process was repeated for all selected points in the RoI, thereby generating the experimental database used in the subsequent analysis.

It is worth noting that the direction-finding stage based on the GLS and WLS estimators does not require the radiometric constant C in (3), since both estimators operate on inter-orientation RSS ratios. However, recovering the full 3D receiver position does require C , as described in [11]. Therefore, an initial calibration step was performed to estimate this constant. Specifically, the PD was positioned directly below the LED at known distances, with the LED pointing vertically downward and the PD pointing upward, so that the resulting RSS measurements could be substituted into (2) to solve for C . This procedure was repeated for receiver heights of 0.4, 0.6, 0.8, 1.0, and 1.2 m. The corresponding estimates

were then averaged, yielding the value $C = 6.2306$, which was subsequently used in the 3D position reconstruction stage.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. SBL Method Validation and Estimator Comparison

This subsection provides the main experimental validation of the proposed 3D SBL positioning method. The two closed-form direction estimators introduced in Section II-B, namely GLS and WLS, are evaluated on the same experimental dataset and under the same $K = 9$ steering configuration. Figs. 2 and 3 illustrate the spatial localization results obtained with GLS, whereas Fig. 4 and Table I summarize the comparative performance of both estimators.

Fig. 2 shows the 3D distribution of the receiver estimates obtained with GLS, with the color map representing the positioning error magnitude. The corresponding x - y and y - z projections are shown in Fig. 3. The first and most important observation is that the SBL method works experimentally in 3D: throughout the measured RoI, the estimated points remain clustered around their corresponding ground-truth positions, despite relying on only one steerable LED and one simple PD receiver. This confirms on real hardware the practical feasibility of the ratio-based SBL principle developed in Section II.

A second observation is that the error is spatially structured rather than uniform. In both the 3D view and the planar projections, the shortest ground-truth-to-estimate displacements are concentrated near the central region of the workspace, while larger deviations appear near the boundaries and corners. This trend is consistent with the geometry of the ratio construction. In the adopted steering set, the reference measurement S_1 corresponds to the nadir-pointing orientation. Hence, for peripheral receiver positions, the ratios $S_{1,k}/S_{1,1}$ involve a denominator acquired under a comparatively large off-axis irradiance angle. Since the LED datasheet specifies a half-power semi-angle of approximately 45° , the reference RSS becomes increasingly sensitive to deviations from the nominal Lambertian law as the receiver moves farther from the beam center. In turn, this sensitivity is propagated and amplified through the ratio-based hyperplane constraints, which explains why the largest errors are predominantly observed at the edges of the testbed.

The quantitative comparison in Table I confirms the superiority of GLS over WLS for all reported metrics. For the 3D positioning error, GLS achieves an RMSE of 12.52 cm, an MAE of 9.97 cm, a $P50$ of 7.93 cm, and a $P90$ of 19.29 cm. In contrast, WLS yields 15.79 cm, 11.98 cm, 8.56 cm, and 27.04 cm, respectively. The gain is especially pronounced in the upper tail of the distribution, where GLS reduces the 90th-percentile error by nearly 8 cm relative to WLS.

This difference is also evident in Fig. 4, where the GLS curve lies consistently to the left of the WLS curve and reaches the 90% probability level at a substantially smaller error. The reason is directly linked to the statistical structure of the SBL ratio model. As discussed in Section II-B, the empirical ratios are not only heteroscedastic but also mutually correlated because they share the same reference orientation.

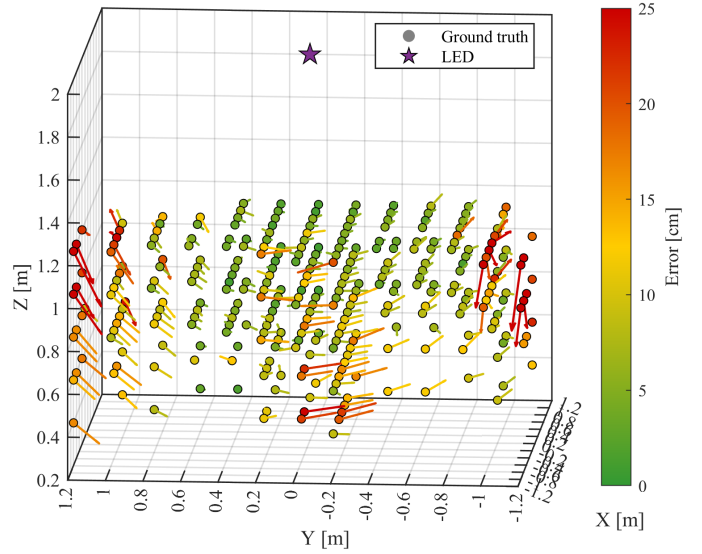


Fig. 2. Experimental 3D positioning results obtained with the GLS estimator for the $K = 9$ SBL configuration. Ground-truth receiver positions are connected to their corresponding estimates, while the color map indicates the magnitude of the 3D positioning error. The LED transmitter is located at $\mathbf{p}_t = [0, 0, 2]^T$ m.

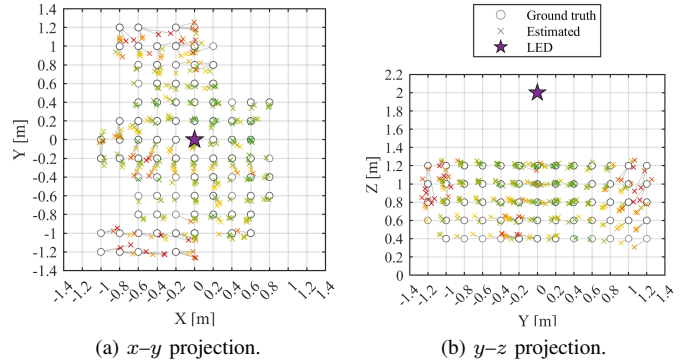


Fig. 3. Planar projections of the experimental localization results obtained with the GLS estimator for the $K = 9$ SBL configuration. The line segments connect each ground-truth receiver position to its corresponding estimate, highlighting the spatial structure of the positioning error across the measured 3D region of interest.

GLS explicitly incorporates the full covariance matrix of the ratio errors, whereas WLS retains only the diagonal weights and neglects cross-correlation terms. Consequently, both estimators behave similarly in favorable regions of the workspace, but GLS is more robust when the ratio reliability becomes uneven across orientations, which is precisely the case for peripheral locations.

Overall, these results establish the central experimental message of the paper: 3D positioning with a single beam-steered LED is feasible in practice using a fully model-based closed-form estimator. Among the two estimators considered here, GLS provides the most accurate and robust performance.

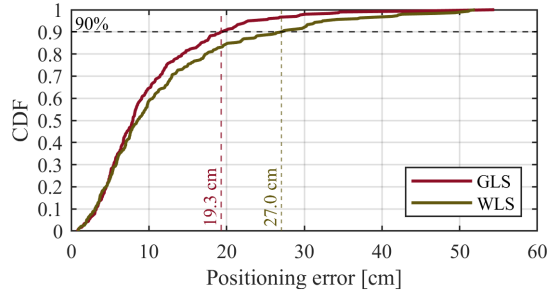


Fig. 4. Empirical CDF of the 3D positioning error for the GLS and WLS estimators under the $K = 9$ SBL configuration.

B. Direction Finding Performance

While the previous subsection assessed the SBL method in terms of full 3D positioning accuracy, for beam-steering applications the complete receiver position is not always required. In many practical optical wireless scenarios, estimating the line-of-sight direction from the transmitter to the user is already sufficient to support beam alignment. For this reason, we now shift the focus from 3D position reconstruction to direction finding.

Let $\mathbf{n}_d$ and $\hat{\mathbf{n}}_d$ denote the ground-truth and estimated direction vectors, respectively. The direction-finding performance is quantified through the pointing error:

$$\alpha = \arccos(\mathbf{n}_d^\top \hat{\mathbf{n}}_d). \quad (16)$$

The aggregate results are reported in Table I. GLS again provides the best performance, with a pointing-error RMSE of 4.87° , and MAE of 4.06° . The corresponding WLS values are 6.15° , 4.87° . Thus, the superiority of GLS is preserved when the evaluation metric is changed from spatial position error to angular beam-pointing error.

This trend is clearly visible in Fig. 5, where the empirical CDF of α is shown for both estimators. The GLS curve lies consistently to the left of the WLS curve and reaches the 90% probability level at a substantially smaller pointing error. From a beam-steering perspective, this is an important result: it shows that the experimentally validated SBL method not only localizes the receiver in 3D, but also estimates the steering direction with single-digit angular accuracy for most measured positions. Since the underlying application is transmitter-side beam alignment, this angular viewpoint is arguably more relevant than full position reconstruction.

To better understand the origin of the residual pointing errors, Fig. 6(c) plots α as a function of the ground-truth inclination θ of the receiver direction. Two regimes can be distinguished. For moderate inclinations, both estimators remain relatively accurate and the pointing error stays low. However, once θ exceeds approximately 40° – 45° , the error starts to increase, and the increase is markedly stronger for WLS than for GLS. This behavior is consistent with the ratio-based formulation, since more oblique directions are associated with weaker and less balanced inter-orientation RSS

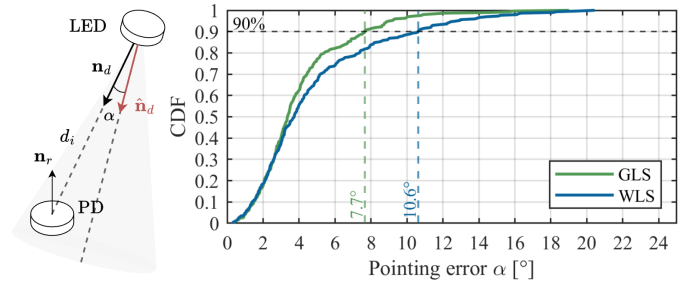


Fig. 5. Empirical CDF of the pointing error for the GLS and WLS estimators. The pointing error is defined as the angular difference between the estimated and true direction of the LED.

TABLE I
EXPERIMENTAL COMPARISON OF THE GLS AND WLS ESTIMATORS IN TERMS OF 3D POSITION AND POINTING ERRORS FOR THE $K = 9$ SBL CONFIGURATION.

	Method	RMSE	MAE	P50	P90
3D position error [cm]	GLS	12.52	9.97	7.93	19.29
	WLS	15.79	11.98	8.56	27.04
Pointing error [$^\circ$]	GLS	4.87	4.06	3.36	7.66
	WLS	6.15	4.87	3.73	10.63

measurements, making the direction estimate more sensitive to model mismatch and ratio uncertainty.

Further insight is provided by Figs. 6(a) and 6(b), which compare the estimated and ground-truth spherical angles of the receiver direction. Fig. 6(a) shows that the main difference between GLS and WLS lies in the inclination estimate. Up to moderate inclination values, both estimators follow the ideal diagonal reasonably well. At larger inclinations, however, the WLS estimates exhibit a stronger systematic deviation and a wider spread, with many samples departing noticeably from the $\pm 5^\circ$ band. In contrast, GLS remains more tightly clustered around the ideal trend, although some bias is still observed at the largest angles. This visual observation is confirmed quantitatively: the inclination RMSE/MAE are $3.29^\circ/2.54^\circ$ for GLS and $4.68^\circ/3.51^\circ$ for WLS. Hence, the main source of the performance gap in pointing error is the improved inclination recovery provided by GLS.

By contrast, Fig. 6(b) shows that the azimuth estimate is accurate for both estimators over almost the entire angular range. Most samples remain close to the ideal diagonal and within the $\pm 10^\circ$ band, with only minor differences between GLS and WLS. This is again supported by the numerical results, since the azimuth RMSE/MAE are $6.88^\circ/5.06^\circ$ for GLS and $6.94^\circ/5.07^\circ$ for WLS. Therefore, the dominant contribution to the overall pointing-error difference does not come from azimuth estimation, but from inclination estimation under increasingly oblique geometries.

Overall, these results show that the proposed SBL method is experimentally effective not only for 3D localization, but also for direct beam-pointing purposes. In that context, direction finding emerges as a particularly meaningful performance metric, and the comparison confirms that GLS provides the most reliable angular estimates, especially in the challenging

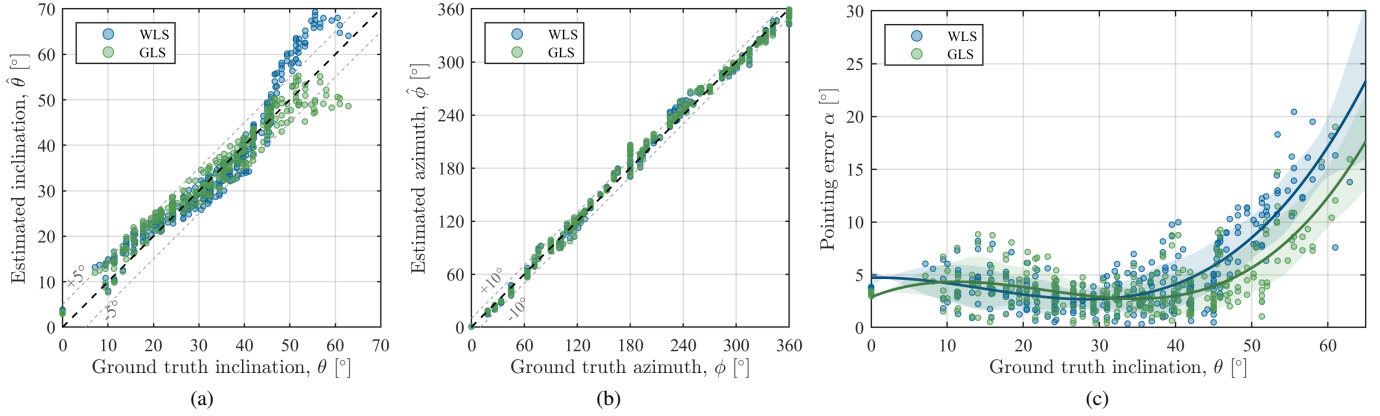


Fig. 6. Direction finding performance of the GLS and WLS estimators: (a) inclination error, (b) azimuth error, and (c) positioning error.

off-axis regime.

C. Position Estimation Stability

Having established GLS as the most reliable estimator, we now examine the stability of the 3D position estimate with respect to the integration time used to average the RSS samples. This analysis is practically relevant because the integration time directly determines the acquisition latency per steering orientation and, therefore, the overall update rate of the SBL system. Since the DAQ operates at 1 kHz, an integration window of T_{int} ms is equivalent to averaging $N = T_{\text{int}}$ samples per orientation.

Fig. 7 reports two complementary metrics as a function of the integration time: the positioning accuracy, quantified by the 3D RMSE with respect to ground truth, and the positioning precision, quantified by the 3D-RMS spread of the estimates. The accuracy is remarkably stable over the entire range from 1 to 1000 ms. The RMSE decreases only marginally. Most of this small improvement is already achieved within the first few tens of milliseconds, after which the curve essentially reaches a plateau. In addition, the precision improves much more noticeably as the integration time increases. The 3D-RMS spread is relatively high for very short windows, around the 10^{-1} cm range at 1 ms, and then drops rapidly as more samples are averaged. By approximately 50 ms, the precision has already fallen below 0.05 cm, and it continues to decrease gradually for longer integration times, reaching the order of 10^{-2} cm for the largest windows. This behavior is consistent with the expected reduction of random noise through temporal averaging.

These results indicate that the dominant limitation of the experimental SBL system is not short-term measurement noise, but rather systematic error sources such as **model mismatch and hardware nonidealities**. Increasing the integration time improves repeatability, but has only a negligible effect on the absolute positioning accuracy. From a system-design perspective, this is an important result: it shows that the proposed method does not require long acquisition windows to preserve its localization performance. In practice, modest integration

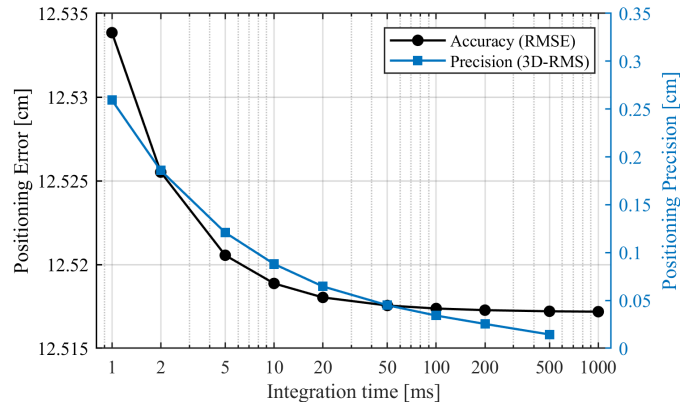


Fig. 7. Effect of the integration time on the GLS-based 3D position estimate for the $K = 9$ SBL configuration. The black curve shows the positioning accuracy in terms of 3D RMSE, while the blue curve shows the positioning precision in terms of 3D-RMS spread.

times on the order of a few tens of milliseconds already provide near-optimal accuracy while significantly reducing the estimation variance, thereby offering a favorable compromise between robustness and latency.

V. CONCLUSIONS

This paper presented, to the best of our knowledge, the first experimental validation of 3D **visible-light positioning** based on a single beam-steered LED and a single **photodiode** receiver. Building on the ratio-based SBL formulation previously derived in [11], the study demonstrated that a fully model-based closed-form approach remains experimentally viable in a real robotic testbed while keeping the receiver architecture extremely simple and concentrating the steering complexity at the infrastructure side. The results showed that the method is effective not only for full 3D localization, but also for direction finding, which is particularly relevant for beam-steering operation in **optical wireless** systems.

Among the two estimators considered, GLS consistently outperformed WLS in both spatial and angular performance.

For 3D positioning, GLS reduced the RMSE from 15.79 cm to 12.52 cm and the $P90$ error from 27.04 cm to 19.29 cm. For direction finding, GLS achieved a pointing-error RMSE of 4.87° and a $P90$ of 7.66° , confirming its superior robustness, especially at large off-axis inclinations. In addition, the integration-time analysis showed that the positioning accuracy remains essentially stable over a wide range of averaging windows, while the estimate precision improves rapidly with modest temporal averaging. Overall, these results validate the practical feasibility of single-LED 3D SBL positioning and support its relevance for future indoor localization and beam-aware optical wireless systems.

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